

Commonality in volatility risk premium

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Abstract

The study examines the existence of commonality in volatility risk premium of National Stock Exchange of India, which is an open electronic limit order book market. The study includes market and stock specific characteristics which may influence the commonality relationship. The key finding of the study is that it produces empirical evidences of commonality in volatility risk premium. The commonality relationship is robust and significant even with the presence of market and stock specific characteristics.

Keywords: Commonality; Model free implied volatility; Realized variance; Volatility risk premium

JEL Classifications: G12; G13; G14

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1. Introduction

The extant literature defines the difference between implied volatility and realized volatility as volatility risk premium (VRP). Despite of the growing body literature on market VRP, a very little is known about the individual stock VRP and its relationship with market VRP. Few studies (e.g. Bakshi and Kapadia, 2003; Carr and Wu, 2008; Cao and Han, 2013; Duan and Wei, 2009; Duarte and Jones, 2007) investigate the pricing mechanism and implications of individual stock VRP. But these studies do not explore the relationship of individual stock VRP to the market VRP.

The main hypothesis of this study is the co-movement of individual stock VRP with the market VRP. The hypothesis is motivated by the recent growing literature on commonality. For example, wide literature has documented significant commonality in liquidity among stocks (Chordia et. al, 2000; Hasbrouck and Seppi, 2001; Huberman and Halka, 2001; Krishnan and Mishra, 2012; Kumar and Shah, 2005). These studies have concluded that liquidity among stocks covary significantly, and there are some common underlying determinants that drive the commonality in the liquidity. In this study, we aspire to investigate the commonality in individual stocks VRP to the market VRP of National Stock Exchange (NSE) of India, which is an open electronic limit order book (LOB) market.

The intuition of the main hypothesis of the study is as follows. As noted by Chordia et. al (2000), trading activity exhibits correlated trading pattern in response to market wide volatility shocks. Volatility shocks induce the investors to create time varying net demand of volatility in options market. Broadly, liquidity demanders of volatility in options market could be the informed players who trade on their private information about volatility. Otherwise, they could be simply hedgers who demand volatility in options market to protect their portfolio under volatility shocks. Liquidity suppliers, on the other hand, supply volatility after covering up their

asymmetric information cost of volatility, apart from all other costs borne by them. Thus, liquidity suppliers mitigate the net demand of market wide volatility by a “volatility markup” process, as described by Green and Figlewski (1999). In this process, liquidity suppliers set option prices such that implied volatility of options exceed the realized volatility rate to cover up all the costs borne by liquidity suppliers. Under the common market wide volatility shocks, the volatility markup process is implemented for the individual stock options as well as for the index options. Thus, plausible reason for commonality in VRP is the “volatility markup” process set by liquidity suppliers in options market because of the market wide volatility shocks. The co-movement or the commonality in VRP has option pricing implications although this study focuses only on the commonality in VRP, which is the main contribution of the study.

We test the commonality hypothesis by employing pooled regression on individual stock VRP with market VRP. We compute individual stock VRP by the difference of model free implied volatility (MFIV) of stocks and realized volatility of stocks. The calculation procedure of MFIV is same as India VIX methodology used by National Stock Exchange (NSE) of India. The MFIV methodology measures the expected volatility of thirty calendar days. Realized volatility is computed by two scaled realized volatility (TSRV) measure for past thirty calendar days. The difference between these two measures is taken as VRP of individual stocks.

Empirically, we find existence of commonality between stock VRP and market VRP. The commonality relationship is robust across the other market specific characteristics and stock specific characteristics. We consider market volatility level, changes of market volatility as market specific characteristics which may influence the commonality relationship. We also consider the stock specific characteristics e.g. stock volatility level, recent changes of stock volatility, firm size, stock liquidity, and stock open interest of put options in the analysis.

Empirical results show both market specific characteristics and stock specific characteristics influence the stock VRP significantly but commonality in VRP results are robust even with the presence of market and stock specific characteristics.

The study proceeds as follows. Section 2 describes the data used in the study. Section 3 discusses the methodology adopted in this study to test our hypothesis. Section 4 discusses the results and main findings of the study. Section 5 concludes the paper.

2. Data

The study uses proprietary NSE data from 30 July 2015 to 30 December 2016. NSE, established in 1994, is anonymous open electronic limit order book market by design and is the largest stock exchange of India in terms of trading volume. The “Futures and Options” and “Equity” segment of the NSE have the same trading hours. Nifty is the leading stock index of NSE consisting of 51 stocks with highest capitalization and liquidity. Nifty options were the highest actively traded index option worldwide in 2015 period. Individual stock options market contribute approximately 6% turnover of the NSE’s derivatives market segment. This study uses 44 individual stocks that comprises Nifty index. These stocks are chosen in terms of their liquidity in options market.

To compute the MFIV of the individual stocks, the study uses derivatives snapshot order book data of NSE at 15:00:00 i.e. thirty minutes before the market close. Order book data of options consists of expiry date, price, quantity, time stamp, buy/sell indicator, book type of every strikes prices of each options traded in the derivatives segment of NSE. To compute the realized volatility of the individual stocks, the study uses equity market trades data provided by NSE. Trades data of NSE consists of time stamp, price, and quantity traded of every symbol traded in spot market of NSE. Study also uses “bhavcopy” data provided by NSE to compute open interest of put options for all the stocks and index used in the research.

3. Methodology

VRP is calculated as the difference between the model free implied volatility (MFIV) to the model free realized volatility. In the subsequent sections, we discuss the calculation procedures of all the variables.

3.1 Model free implied volatility of individual stocks

Calculation procedure of India VIX by NSE is employed to compute individual stock option's MFIV. India VIX computation procedure represents the MFIV methodology. India VIX follows the Chicago Board of Options Exchange (CBOE) computation methodology with suitable adjustments to acclimate Nifty Options order book. In the similar way to India VIX[†] methodology, we compute the individual stock option's MFIV taking the best bid-ask scores of near and next month contracts that are traded in the F&O segment of NSE. The following formula is used to calculate MFIV of individual stock options.

$$stock_mfiv_{j,t}^2 = \frac{2}{T} \sum_i \frac{\Delta K}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$$

$stock_mfiv_{j,t}$ = MFIV of jth stock options

T = Time to expiration

F = Forward level of individual stock as the latest available price of jth stocks future contract of the corresponding expiry

K_0 = First strike below the forward level

K_i = Strike price of ith out-of-money option; a call if $K_i > F$ and a put if $K_i < F$

ΔK_i = Interval between strike prices i.e. half the distance between the strikes on either side of K_i (For the lowest strike, ΔK_i is the difference between lowest strike and the next higher strike. Similarly, for the highest strike, ΔK_i is the difference between highest strike and the next lower strike)

R = Risk free interest rate to corresponding expiration

$Q(K_i)$ = Midpoint of the bid-ask quote for each option contract with strike K_i

[†] https://www.nseindia.com/content/indices/white_paper_IndiaVIX.pdf

There are two differences between CBOE and NSE computation of MFIV. CBOE computes forward level by the put call parity relation at a strike where the absolute difference between call and put prices is the smallest. NSE computes MFIV by taking latest available futures price of the underlying. Second, CBOE uses observed midpoint to compute $Q(K_i)$. But liquidity of far month traded options is a problem for Indian market. Thus, a natural cubic spline interpolation is used to fill the midpoint prices where the bid-ask spread is more than thirty percent. Additionally, to handle liquidity issues individual options, we follow Grover and Thomas (2012) and assign zero weight for those strikes where bid or ask or both do not exist. Further, we compute the daily MFIV of individual stocks at the 15:00:00 o'clock (thirty minutes before market closing) from the NSE order book. Apart from individual stock MFIV, we use closing value of the India VIX provided by NSE as market index MFIV.

3.2 Model free realized volatility of individual stocks

Two scaled realized volatility (TSRV) is employed as the model free realized volatility in the study. We follow Vipul and Jacob (2007) to calculate TSRV, which is as follows.

$$stock_rv_{j,t}^2 = \frac{N}{(N-\bar{n})} [\bar{\sigma}_{lowfrequency,t}^2 - \frac{\bar{n}}{N} \sigma_{highfrequency,t}^2]$$

$stock_rv_{j,t}$ = model free realized volatility measure of j^{th} stock

$\bar{n}$ = average number of returns across subsamples at the low frequency

N = Total number of returns at the high frequency

$\bar{\sigma}_{lowfrequency,t}^2$ = average variance at low frequency; low frequency is calculated with five minutes

$\sigma_{highfrequency,t}^2$ = variance at high frequency; high frequency is calculated with one second

The above measure of realized volatility calculates only open market volatility. In order to calculate realized volatility for the entire day, we need to calculate close market volatility.

Thus, we scale up the above mentioned volatility by the ratio of daily close-to-close to daily open-to-close variances following Koopman et al.(2005), Garg and Vipul (2015) . The scaling factor ρ_j for j^{th} stock is calculated as mentioned below.

$$\rho_j = \frac{\sum_{t=1}^T r_{cc,j}^2}{\sum_{t=1}^T r_{oc,j}^2}$$

$r_{cc,j}^2$ =daily close-to-close return of the j^{th} stock

$r_{oc,j}^2$ =daily open-to-close return of the j^{th} stock

We employ above mentioned TSRV measure to calculate both individual stock and market index (Nifty) VRP.

3.3 Volatility risk premium

The traditional definition of VRP is the difference between the expected risk neutral measure of volatility and expected objective measure of volatility.

$$VRP_{t=} E_t^Q \left(Var_{t,t+1} \right) - E_t^P (Var_{t,t+1})$$

We calculate MFIV of individual stocks as a measure of expected risk neutral volatility. The MFIV measure calculated by the above procedure is the expected risk neutral volatility of the next thirty calendar days. Thus, the next focus is to forecast the realized volatility (calculated from the scaled TSRV measure) to the next thirty calendar days. In Indian market, Garg and Vipul(2014) document that for the monthly forecast, random walk model dominates the other forecasting models (Like EWMA, ARFIMA, HAR) based on the efficiency and bias criteria. Based on the above findings we use random walk model to forecast thirty calendar days realized volatility, which would be comparable to the MFIV measure. Under the random walk model measure, the scaled TSRV values are added up to for past thirty calendar days, and next thirty calendar days forecast is made by the random walk model specified below.

$$\hat{\sigma}_t = \sigma_{t-1}$$

σ_{t-1} =estimate of the volatility at time t-1

$\hat{\sigma}_t$ =forecast of volatility at time t

We calculate VRP for individual stocks and market index as specified below. The daily market VRP is calculated based on the difference of daily closing India VIX value and forecasted realized volatility (scaled TSRV as mentioned above) of Nifty. Similarly, the individual stock VRP is measured as the difference between calculated MFIV and forecasted realized volatility for each individual stocks.

$$\begin{aligned} \text{Market_VRP}_t &= \text{IVIX}_t - \text{RV}_{t,t-30} \\ \text{stock_vrp}_{j,t} &= \text{stock_mfiv}_{j,t} - \text{stock_rv}_{j,t} \end{aligned}$$

3.4 Market and stock specific characteristics

In the study, we control for market specific and stock specific characteristics, which we discuss below.

3.4.1 Market specific characteristics

In the market specific factors, we control for the market volatility and change of the volatility. The intuition behind controlling for the market volatility is that the level of volatility affects the VRP of the stocks. Volatility of an asset is mean reverting in nature so as market volatility. As trading activity exhibits correlated trading patterns in response to market wide volatility shocks (Chordia et. al, 2000), high (low) level of market volatility would induce individual stocks to exhibit high (low) level volatility. When volatility level is high (low), the “volatility markup” process would yield lower (higher) VRP. This is because VRP is the difference between implied and realized volatility and when realized volatility level is high (low), the difference of these two volatility levels would decrease (increase). Thus, “volatility markup”

process that sets implied volatility over and above realized volatility would yield lower (higher) VRP in times of high (low) volatility levels. The individual stock VRP and market volatility level exhibits negative relationship if the above intuition holds true. We calculate daily level of volatility by the difference of Nifty's intraday high and low price divided by the closing Nifty price of the day.

We also control for the change of the market volatility following Goyal and Saretto (2009). Although Goyal and Saretto (2009) explain that deviations of implied volatility from realized volatility are the signs of volatility mispricing, we have different explanation for the deviation. We argue that liquidity suppliers set option prices such that implied volatility exceeds the realized volatility rate to cover up all the costs borne by the liquidity suppliers. According to Stein (1989) and Poteshman (2001), investors overreacts (underreacts) to recent high (low) changes of volatility and pay high (low) premium for high (low) volatility shocks. We differ from this set of explanation and we argue that liquidity suppliers set higher (lower) implied volatility to the recent high (low) change of volatility. Thus, the level of VRP (difference between implied and realized volatility) would be positive related to the recent changes of the volatility. We compute changes of recent changes of volatility by the difference between today's volatility and previous trading day's volatility scaled by the previous trading day's volatility.

3.4.2 Firm specific characteristics

Similar to the market volatility level and recent changes of market volatility, we control for the individual stocks volatility level and recent changes of individual stocks volatility. As argued above (Chordia et. al, 2000), correlated trading patterns would induce high (low) volatility to the individual stocks in tandem to market volatility. Thus, the control for stocks volatility and changes of volatility become necessary. Like market volatility level, stocks volatility levels

should exhibit negative relationship with the stocks VRP level. Additionally, the recent changes of volatility should exhibit positive effect with stock VRP for the similar arguments discussed above.

Apart from the level of volatility and changes of volatility, we also control for the firm size and stock liquidity. Firm size and firm liquidity may have effect on VRP of the stocks. It is a common belief that information asymmetry is lower for higher sized firms because of high analyst coverage and forecast. As VRP is a measure of option's expensiveness, which is related to information asymmetry that the liquidity supplier assumes, would be a function of size of the firm. The larger the size of the firm, lower would be the VRP because of less information asymmetry. Stock liquidity might also have effect on VRP. Highly liquid firms often enjoy less information asymmetry because of the high trading volumes. Whereas, less liquid firms often suffers from more information asymmetry. Thus, liquidity suppliers would tend to set higher (lower) volatility markup for less (high) liquidity stocks because of higher (lower) information asymmetry. We calculate the size of a firm by the total number of equity outstanding for the stock multiplied by the NSE closing prices. We use natural logarithm of the size as our independent variable in the main regression. Additionally, we also take squared size of a firm as our control variable assuming there may exist a non-linear relationship between size of the firm and stock VRP.

We include daily open interest of put options as a proxy for option demand pressure. Previous studies emphasize the role of constrained financial intermediaries (Cao and Han, 2013; Garleanu et. al, 2009; Fan et. al, 2016) in imperfect market. Studies document that option prices are affected by the demand of the options (Bollen and Whaley, 2004; Garleanu et. al, 2009) and when there are limits to arbitrage (Shleifer and Vishny, 1997), because of costly hedge or replication of the option, liquidity suppliers supply options with higher prices. Thus, option demand affects option prices positively. Put options provide investors to hedge against

volatility crash. Thus, open interest of put option provides a natural proxy for hedging volatility demand. We include the daily open interest of put options for individual stocks as a proxy for tail risk demand.

3.5 Descriptive statistics

Descriptive statistics of the stock VRP and market specific characteristics are reported in Table 1. The mean of average stock VRP is noticeably negative for the period of study where as mean of market VRP is positive. Further, the standard deviation of the average stock VRP is very high compared to the market VRP. All the variables are stationary.

Figure 1 plots the average market VRP is considerably lower than market VRP. Further, average market VRP is negative in most of the time periods. Descriptive statistics of the stock specific characteristics are reported in Table 2. Noticeably, all the variables in Table 2 are stationary except firm size. Although the variable is not stationary, we do not detrend the variable following Lo and Wang (2000), where they advised to take shorter period of time (typically five years) as detrend cannot be achieved without removing adequate serial correlation. In the descriptive statistics the Jarque-Bera test has a null hypothesis of normality and ADF test has null hypothesis of stationarity. We report the t -statistics of these tests.

Table 1: Descriptive statistics of the stock VRP and market specific characteristics.

Below table reports the descriptive statistics of average stock VRP, market VRP, market volatility level and change of market volatility. Jarque-Bera test has a null hypothesis of normality. ADF test has null hypothesis of stationarity.

	$stock_vrp_{j,t}$	$Market_VRP_t$	$Market_Volatility_t$	$\Delta Market_Volatility_t$
Mean	-0.093	1.255	1.141	0.167
Standard Deviation	1.346	0.631	0.621	0.718
Skewness	-0.745	-0.202	2.449	2.111
Kurtosis	3.113	4.187	13.045	10.408
Jarque-Bera (<i>t</i> -statistics)	31.611	22.283	1769.088	1030.060
ADF test (<i>t</i> -statistics)	-5.440	-5.228	-6.949	-17.705
#Obs	340	340	340	340

Figure 1: Plot of average stock VRP and market VRP

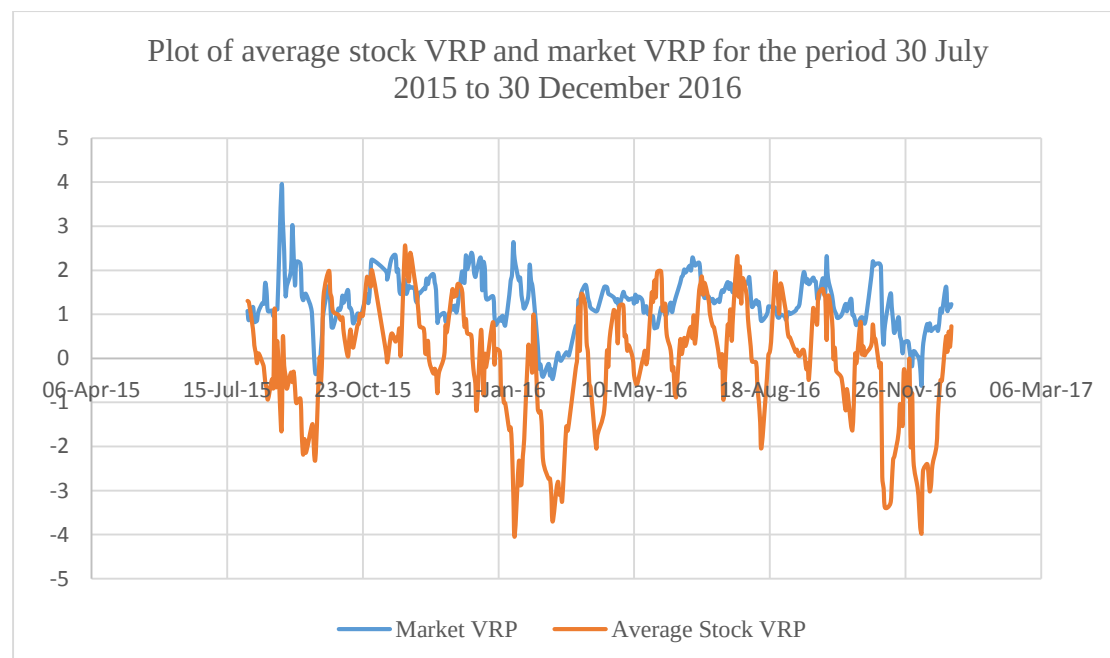


Table 2: Descriptive statistics of the stock specific characteristics.

Below table reports the descriptive statistics of average stock volatility, average changes of stock volatility, average firm size, average firm liquidity and average open interest of put options of individual stocks. Jarque-Bera test has a null hypothesis of normality. ADF test has null hypothesis of stationarity

	$stock_volatility_{j,t}$	$\Delta stock_volatility_{j,t}$	$stock_size_{j,t}$	$stock_liquidity_{j,t}$	$stock_putoi_{j,t}$
Mean	2.625	0.147	10.247	14.388	13.447
Standard Deviation	0.808	0.299	0.101	0.268	0.370
Skewness	2.889	2.884	0.266	0.659	-0.584
Kurtosis	17.372	18.473	2.812	5.278	3.173
Jarque-Bera (t-statistics)	3398	3863	4.519	98.173	19.741
ADF test (t-statistics)	-5.333	-24.245	-1.450	-7.015	-7.684
# Obs	340	340	340	340	340

3.6 Main regression equation

We employ the following econometric specification to regress the stock VRP on the market VRP and other market and stock specific characteristics to understand the extent to which individual stock VRP covary with market VRP. The general regression specification is

$$\begin{aligned}
 stock_vrp_{j,t} = & a + \sum_{i=-1}^{+1} b_{i+1} Market_VRP_{t+i} + \sum_{k=-1}^{+1} c_{k+1} Market_Volatility_{t+k} + \\
 & d_1 \Delta Market_Volatility_t + e_1 stock_volatility_{j,t} + f_1 \Delta stock_volatility_{j,t} + \\
 & g_1 stock_size_{j,t} + h_1 stock_size_{j,t}^2 + m_1 stock_liquidity_{j,t} + n_1 stock_putoi_{j,t} + \varepsilon_t \quad (1)
 \end{aligned}$$

We perform pooled regressions and compute t -statistics from Newey-West (1987) standard error that correct for cross sectional correlations in the data. $stock_vrp_{j,t}$ represents VRP of stock j at time t . We compute the daily VRP of stock j thirty minutes before the market

closing. Similarly, $Market_VRP_t$ represents the market (index Nifty) VRP at time t . Market volatility level is represented as $Market_Volatility_t$ and change of market volatility as $\Delta Market_Volatility_t$. We include stock specific factors such as $stock_volatility_{j,t}$ and $\Delta stock_volatility_{j,t}$ that represent the individual stock daily volatility level and change of volatility of stock j at time t . We include firm specific factors like size of the firm and squared stock size in our regression equation as $stock_size_{j,t}$ and $stock_size_{j,t}^2$. Similar to firm size, we include stock liquidity of stock j at time t as $stock_liquidity_{j,t}$. Consistent with the demand hypothesis of the options we include open interest of put options of stock j at time t as $stock_putoi_{j,t}$ as the representative of hedging tail demand.

We include lead and lag terms of market VRP to capture the lagged adjustments of commonality in VRP because of non-synchronous trading. Existence of the commonality in the VRP between individual stock and market would mean that any of the b_{i+1} coefficients would be statistically significant and positive. We also include lead and lag terms of market volatility as control for the commonality relationship between stock and market VRP. Apart from commonality, we also test the stock VRP behaviour based on the size of the firm of stock j and also on the hedging demand of options by the open interest of put options on stock j .

4. Results and Discussion

We present potential explanations of our results in this section. We first discuss the results of commonality and market specific factors as described in section three. Then we discuss the results of commonality in VRP taking stocks specific factors. Lastly, we discuss the overall results of the model.

4.1 Commonality with market specific factors

First, we examine the commonality between stock VRP and market VRP in Table 3. Table 3 shows that the relationship between individual stock VRP and market VRP is significantly positive for all the models. The significant positive relationship is robust even with market specific factors like market volatility and change of market volatility. Rather, in the presence of market volatility and change of market volatility factors (Model 3), the estimated contemporaneous coefficient of market VRP is 0.491 with t -statistics of 10.51, which shows contemporaneous market VRP is the most significant factor (in terms of magnitude and significance) in movement of stock VRP among the lag, contemporaneous and lead market VRP. Additionally, all the lead and lag coefficients of market VRP are significantly positive for all the models. Lag and lead of market VRP are included to capture lagged adjustments in commonality in VRP because of non-synchronous trading. The positive significant relationship between stock VRP and market VRP is inferred as the evidence of existence of commonality in VRP.

The relationship between stock VRP and market volatility is significantly negative in Model 1 and Model 2. In fact, contemporaneous market volatility shows the most negative significant relationship with coefficient -0.958 with t -statistics -9.42 in Model 3. Moreover, the lag and lead market volatility coefficients are also negatively significant. That shows commonality in VRP depends upon the level of market volatility. In times of high (low) volatility level, the “volatility markup” process yields low (high) level of stock VRP because of the reason that the difference between MFIV and realized volatility decreases (increases) in time high (low) realized volatility.

Change of market volatility is significantly positive with stock VRP with coefficient 0.323 and t -statistics 5.16 in Model 3. The recent change of market volatility induces liquidity

suppliers to set option prices in tandem to recent change of volatility. Thus, higher (lower) changes of recent volatility induces liquidity suppliers to keep option prices high (low).

Table 3: Commonality with market specific factors

This table reports the commonality in stock and market VRP with market specific factors. Daily stock and market VRP are calculated by the difference of MFIV and realized volatility measured from scaled TSRV. Market volatility for a day is measured by the Nifty's intraday high and low price divided by the closing Nifty price. Change of the market volatility is measured by the change of volatility in two consecutive trading days scaled by the previous day's volatility. The sample period is from 30 July 2015 to 30 December 2016. Robust Newey and West (1987) t -statistics are reported in the brackets. *, **, *** denote the statistical significance at 1%, 5%, and 10% levels respectively

Variables	Model 1	Model 2	Model 3
Intercept	-1.057*** (-5.68)	0.242 (1.25)	0.235 (1.22)
$Market_VRP_{t-1}$	0.164** (2.43)	0.329*** (4.57)	0.314*** (4.37)
$Market_VRP_t$	0.243*** (4.92)	0.458*** (10.02)	0.491*** (10.51)
$Market_VRP_{t+1}$	0.418*** (6.67)	0.237*** (3.87)	0.223*** (3.63)
$Market_Volatility_{t-1}$		-0.405*** (-6.62)	-0.148*** (-2.7)
$Market_Volatility_t$		-0.642*** (-11.47)	-0.958*** (-9.42)
$Market_Volatility_{t+1}$		-0.315*** (-5.8)	-0.301 (-5.67)
$\Delta Market_Volatility_t$			0.323 (5.16)
Average Adjusted R^2	0.0225	0.0572	0.0581

4.2 Commonality with stock specific factors:

Next we study how the commonality relationship is affected by the stock specific characteristics. Table 4 reports the results of pooled regression of stock specific characteristics. Table 4 shows that for all the models, all the market VRP coefficients (lag, contemporaneous and lag) are all positively significant to the stock VRP. Thus, commonality relationship does

not get affected by the presence of the stock specific factors. Interestingly, the *t*-statistics of contemporaneous market VRP is 7.19 with coefficient of 0.357 in the Model 7. This *t*-statistics is the highest among all the variables in Model 7 indicating that commonality in VRP is statistically most significant variable.

First, we include stock volatility level in the commonality relationship. Model 1 shows that like market volatility level, stock volatility level is significantly negative. This result may be explained by the similar arguments of market volatility level. “Volatility markup” in which liquidity suppliers set stock option prices to exceed realized volatility rate, yields low (high) VRP level in times of high (low) stock volatility level since VRP level of the stocks is the difference between implied and realized volatility level. When we include recent changes of stock volatility in the regression (Model 2), the coefficient of changes of stock volatility is significantly positive without affecting commonality or stock volatility level results. This result indicate that recent changes in stock volatility affects the stock VRP level. High (low) changes of recent changes of volatility prompts the liquidity suppliers to set option prices higher (lower). This result is also consistent with the recent changes of market volatility.

We include firm size in the commonality relationship as shown in Model 3. The result shows that firm size is negatively significant with stock VRP level. The intuition is, larger the size of the firm, greater the analyst coverage and lower the information asymmetry. As VRP level can be thought as the proxy for option’s expensiveness, lower (higher) the information asymmetry of a firm produces lesser (higher) expensiveness of the options. This is because, liquidity suppliers set lower (higher) prices of the options with the lower (higher) information asymmetry involved with the firm. When we include quadratic term of the firm size in the pooled regression, it is coming positively significant. Thus, non-linear relationship exists between the stock VRP level and firm size. Results of pooled regression with firm size and

quadratic firm size indicate that there exists a U-shaped relationship between firm size and stock VRP level.

Next, we include stock liquidity in the commonality relationship. Inclusion of the stock liquidity does not change the commonality relationship. The coefficient of stock liquidity is negative although not statistically significant. This result is quite intuitive. Higher (lower) traded firms will have lower (higher) trading costs (Forster and Viswanathan ,1993). Thus, liquidity suppliers suffer from lesser (higher) price impact for higher (lower) traded firms prompting them to set option prices lower (higher). Accordingly, stock liquidity displays negative relationship with stock VRP. When we include firm size and its quadratic variation along with stock liquidity in Model 5, the commonality relationship does not change. All the coefficients are consistent with their expected signs.

We include open interest of put options in the commonality relationship in Model 6. Open interest of put options is a natural proxy of option demand for tail risk and volatility shocks. Constrained financial intermediaries (Bollen and Whaley, 2004; Garleanu et. al, 2009) affect the option prices conditional on demand of options. Because of limits to arbitrage, liquidity suppliers of options set prices according to the demand of the options. Conditional on high (low) demand of options, liquidity suppliers supply options making option prices to be high (low). Thus, the stock VRP (difference between MFIV and realized volatility) varies positively with the option demand. Results of Model 6 indicate that open interest of put options positively affects stocks VRP with statistical significance. This result is consistent with Fan et. al, (2016). In Model 7, we include all the stock specific characteristics. The results are consistent with no meaningful change to any of the stock specific variables as discussed above.

Table 4: Commonality with stock specific factors:

This table reports the commonality in stock and market VRP with stock specific factors. Daily stock and market VRP are calculated by the difference of MFIV and realized volatility measured from scaled TSRV. Individual stock volatility for a day is measured by the stock's intraday high and low price divided by the closing stock price. Change of the stock volatility is measured by the change of volatility in two consecutive trading days scaled by the previous day's volatility. Stock size is calculated by the total number of equity outstanding for the stock multiplied by the NSE closing prices for that day. We take natural logarithm of the size in the regression. The liquidity of a stock is measured by the total traded value of the stock for that day. Stock put open interest is the total number of open interest of put options outstanding for the day. The sample period is from 30 July 2015 to 30 December 2016. Robust Newey and West (1987) t -statistics are reported in the brackets. *, **, *** denote the statistical significance at 1%, 5%, and 10% levels respectively

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Intercept	-0.404** (-2.2)	-0.318* (-1.65)	9.635*** (2.71)	-0.766 (-1.22)	10.101*** (2.74)	-2.436*** (-4.32)	8.568** (2.4)
$Market_VRP_{t-1}$	0.189*** (2.83)	0.229*** (3.33)	0.162** (2.42)	0.164** (2.44)	0.163** (2.42)	0.165** (2.44)	0.240*** (3.45)
$Market_VRP_t$	0.383*** (7.63)	0.351*** (7.11)	0.240*** (4.87)	0.243*** (4.94)	0.241*** (4.89)	0.242*** (4.92)	0.357*** (7.19)
$Market_VRP_{t+1}$	0.324*** (5.5)	0.325*** (5.51)	0.421*** (6.74)	0.416*** (6.69)	0.419*** (6.76)	0.416*** (6.62)	0.306*** (5.19)
$stock_volatility_{j,t}$	-0.284*** (-6.03)	-0.331*** (-5.58)					-0.360*** (-5.49)
$\Delta stock_volatility_{j,t}$		0.192** (2.2)					0.221** (2.27)
$stock_size_{j,t}$			-1.974*** (-2.8)		-1.999*** (-2.82)		-1.809*** (-2.63)
$stock_size_{j,t}^2$			0.089*** (2.59)		0.090*** (2.62)		0.078** (2.32)
$stock_liquidity_{j,t}$				-0.020 (-0.46)	-0.024 (-0.54)		-0.131* (-1.71)
$stock_putoi_{j,t}$						0.103*** (2.58)	0.241*** (3.72)
Average Adjusted R^2	0.0386	0.0406	0.0297	0.0225	0.0297	0.0248	0.0565

4.3 Commonality with market and stock specific factors

Table 5 reports the commonality results with all the market and stock specific characteristics. Inclusion of all the market and stock specific variables does not change the commonality relationship between stock and market VRP. All the market VRP coefficients are significantly

positive, contemporaneous market VRP being the most important in terms of both magnitude and statistical significance. The contemporaneous market VRP coefficient is 0.490 with t -statistics being 10.59. All the market specific characteristics are consistent with the results of commonality with market factors as discussed in section 4.1. All the commonality and market factors are statistically significant with expected signs. Likewise, stock specific characteristics are also statistically significant (except stock liquidity) and consistent with the expected signs. There is no meaningful change of the results of the stock characteristics as discussed in section 4.2.

Overall, all the results are the evidences of the existence of commonality in VRP. This relationship is robust with all the market and stock specific characteristics. The commonality in the VRP signifies the correlated market wide trading activity, specifically in terms of volatility pricing activity by the liquidity suppliers of the options. We find stock VRP has significant negative relationship with market volatility level and significant positive relationship with recent changes of market volatility. Thus, market wide volatility level and market wide volatility shocks influence the stock option suppliers to set the option prices accordingly. Moreover, similar to market volatility level and recent changes of market volatility, individual stock volatility level and recent changes of stock volatility retain significant negative and positive relationship respectively. Apart from that, results show that stock VRP maintains a significant U-shaped behaviour with firm size. Stock VRP is also negatively related to the stock liquidity although the relationship is hardly statistically significant. Consistent with the constrained intermediaries hypothesis, we also find open interest of put options, which is a proxy for option demand, has significant positive relationship with the stock VRP.

Table 5: Commonality with market and stock specific factors:

This table reports the commonality in stock and market VRP with stock specific factors. Daily stock and market VRP are calculated by the difference of MFIV and realized volatility measured from scaled TSRV. Market volatility for a day is measured by the Nifty's intraday high and low price divided by the closing Nifty price. Change of the market volatility is measured by the change of volatility in two consecutive trading days scaled by the previous day's volatility. Individual stock volatility for a day is measured by the stock's intraday high and low price divided by the closing stock price. Change of the stock volatility is measured by the change of volatility in two consecutive trading days scaled by the previous day's volatility. Stock size is calculated by the total number of equity outstanding for the stock multiplied by the NSE closing prices for that day. We take natural logarithm of the size in the regression. The liquidity of a stock is measured by the total traded value of the stock for that day. Stock put open interest is the total number of open interest of put options outstanding for the day. The sample period is from 30 July 2015 to 30 December 2016. Robust Newey and West (1987) t -statistics are reported in the brackets. *, **, *** denote the statistical significance at 1%, 5%, and 10% levels respectively

Variables	Overall model
Intercept	9.990*** (2.79)
$Market_VRP_{t-1}$	0.321*** (4.45)
$Market_VRP_t$	0.490*** (10.59)
$Market_VRP_{t+1}$	0.214*** (3.48)
$Market_Volatility_{t-1}$	-0.061 (-1.13)
$Market_Volatility_t$	-0.797*** (-8.67)
$Market_Volatility_{t+1}$	-0.262*** (-5.31)
$\Delta Market_Volatility_t$	0.321*** (5.27)
$stock_volatility_{j,t}$	-0.189*** (-3.23)
$\Delta stock_volatility_{j,t}$	0.131* (1.8)
$stock_size_{j,t}$	-1.891*** (-2.74)
$stock_size_{j,t}^2$	0.083** (2.46)
$stock_liquidity_{j,t}$	-0.123 (-1.6)
$stock_putoi_{j,t}$	0.199*** (3.09)
Average Adjusted R^2	0.0739

5. Conclusion

The paper examines the existence of commonality in stock VRP and market VRP. Previous studies document the correlated trading activity in stock market by examining commonality in liquidity. Correspondingly, this study aims to understand the correlated trading activity both in stock and options market by examining commonality in VRP, which is a derived measure from both spot and options market.

The key finding of the study is that it produces evidences of the commonality in VRP. Further, the relationship is robust across other control variables. We include market specific and stock specific factors that may influence the commonality relationship. Our findings suggest that commonality relationship between stock and market VRP is robust and significant both statistically and economically, even in the presence of all the other factors.

Our key finding is consistent with the market specific factors. In market specific factors, we find market volatility affects the stock VRP negatively and significantly. Moreover, we find positive significant relationship between the recent changes in market volatility and stock VRP. Among stock specific factors, similar to market volatility and changes of market volatility, we find significant negative relationship with stock volatility level and significant positive relationship with recent changes of stock volatility. Firm size shows significant U-shaped behaviour with the stock VRP. Stock liquidity affects the stock VRP negatively. Further, consistent with the constrained financial intermediaries hypothesis, our results show that option demand affects the stock VRP significantly and positively. Our overall results suggest that commonality in stock and market VRP exists and the relationship is robust and significant.

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